

FitTrack360 Database Design

Tables, Attributes, Purpose, and Key Relationships

PK = Primary Key
FK = Foreign Key



1 USER MANAGEMENT



roles

Purpose: Stores Admin, Trainer, Member, Guardian roles.

- role_id (PK)
- role_name



age_groups

Purpose: Stores Child, Youngster, Senior age categories.

- age_group_id (PK)
- group_name
- min_age
- max_age



users

Purpose: Stores user account details and links to role and age group.

- user_id (PK)
- full_name
- email
- password_hash
- dob
- gender
- role_id (FK -> roles.role_id)
- age_group_id (FK -> age_groups.age_group_id)
- guardian_user_id (FK -> users.user_id, optional)
- is_active
- created_at



2 SUBSCRIPTION MANAGEMENT



subscription_plans

Purpose: Stores free and paid subscription plans.

- plan_id (PK)
- plan_name
- price
- duration_days
- plan_type
- description



user_subscriptions

Purpose: Stores each user's subscription history and current status.

- subscription_id (PK)
- user_id (FK -> users.user_id)
- plan_id (FK -> subscription_plans.plan_id)
- start_tate
- end_date
- status



3 FITNESS MANAGEMENT



exercise_categories

Purpose: Stores exercise types like Yoga, Cardio, Strength, Meditation.

- category_id (PK)
- category_name



exercises

Purpose: Stores exercise details.

- exercise_id (PK)
- exercise_name
- category_id (FK -> exercise_categories.category_id)
- difficulty
- suitable_for_age_group_id (FK -> age_groups.age_group_id)
- duration_minutes
- calories_burned
- is_premium
- instructions



workout_plans

Purpose: Stores trainer-created fitness plans.

- workout_plan_id (PK)
- plan_name
- trainer_id (FK -> users.user_id)
- age_group_id (FK -> age_groups.age_group_id)
- difficulty
- is_premium
- created_at



plan_exercises

Purpose: Junction table storing exercises included in a workout plan.

- plan_exercise_id (PK)
- workout_plan_id (FK -> workout_plans.workout_plan_id)
- exercise_id (FK -> exercises.exercise_id)
- sequence_no



workout_sessions

Purpose: Stores completed workout/exercise logs.

- session_id (PK)
- user_id (FK -> users.user_id)
- exercise_id (FK -> exercises.exercise_id)
- session_date
- duration_minutes
- calories_burned



user_plan_enrollments

Purpose: Stores which user enrolled in which workout plan.

- enrollment_id (PK)
- user_id (FK -> users.user_id)
- workout_plan_id (FK -> workout_plans.workout_plan_id)
- enrollment_date
- status



4 HEALTH TRACKING



body_measurements

Purpose: Stores height and weight history.

- measurement_id (PK)
- user_id (FK -> users.user_id)
- height_cm
- weight_kg
- recorded_date



meal_logs

Purpose: Stores user-calorie intake / meals consumed.

- meal_log_id (PK)
- user_id (FK -> users.user_id)
- food_id (FK -> food_items.food_id)
- meal_type
- servings
- log_date



water_logs

Purpose: Stores daily water intake.

- water_log_id (PK)
- user_id (FK -> users.user_id)
- intake_ml
- log_date



food_items

Purpose: Stores food nutrition and calorie details.

- food_id (PK)
- food_name
- calories_per_serving
- protein_g
- carbs_g
- fat_g



medications

Purpose: Stores medicines assigned to users.

- medication_id (PK)
- user_id (FK -> users.user_id)
- medicine_name
- dosage
- frequency
- start_date
- end_date



medication_logs

Purpose: Stores medicine taken/missed history.

- medication_log_id (PK)
- medication_id (FK -> medications.medication_id)
- taken_date
- status



KEY RELATIONSHIPS

#	Relationship	Type	Description	#	Relationship	Type	Description
1	roles -> users	One-to-Many	One role can be assigned to many users.	11	users -> user_plan_enrollments	One-to-Many	One user can enroll in many plans.
2	age_groups -> users	One-to-Many	One age group can have many users.	12	workout_plans -> user_plan_enrollments	One-to-Many	One workout plan can have many enrolled users.
3	users -> user_subscriptions	One-to-Many	One user can have many subscription records.	13	users -> workout_sessions	One-to-Many	One user can complete many workout sessions.
4	subscription_plans -> user_subscriptions	One-to-Many	One plan can be subscribed by many users.	14	exercises -> workout_sessions	One-to-Many	One exercise can appear in many completed sessions.
5	exercise_categories -> exercises	One-to-Many	One category can have many exercises.	15	users -> body_measurements	One-to-Many	One user can have multiple measurement records.
6	age_groups -> exercises	One-to-Many	Exercises can be targeted to a specific age group.	16	users -> meal_logs	One-to-Many	One user can log many meals.
7	users (trainer) -> workout_plans	One-to-Many	One trainer can create many workout plans.	17	food_items -> meal_logs	One-to-Many	One food item can appear in many meal logs.
8	age_groups -> workout_plans	One-to-Many	One age group can have many workout plans.	18	users -> water_logs	One-to-Many	One user can have many water logs.
9	workout_plans -> plan_exercises	One-to-Many	One workout plan contains many plan_exercise rows.	19	users -> medications	One-to-Many	One user can have many medications.
10	exercises -> plan_exercises	One-to-Many	One exercise can appear in many workout plans.	20	medications -> medication_logs	One-to-Many	One medication can have many medication status logs.



TIP: All relationships are enforced through Foreign Keys (FK). Primary Keys (PK) uniquely identify each record in a table.